

Your notes are usable for revision, but not enough for interviews. Interviewers do not want definitions only. They want: **what it is, where it is used, why it matters, one example, and how you would verify it in real life**. Also, a few parts in your list are sloppy or slightly wrong, and that will hurt you.

First, the corrections you need to know

1) “Hub, switch, router”

A hub is basically dead technology. Mention it only as a contrast point. Real interviews care more about **switch vs router**.

2) OSPF area types

“Normal area” is not really a special OSPF type. The correct useful types are: **backbone area (0), normal area, stub, totally stubby, NSSA**. Also know **ABR** and **ASBR**.

3) MPLS

Do not say MPLS is “faster” as a blanket statement. The real value is **label-based forwarding, traffic engineering, VPN separation, and scalable WAN design**. Raw speed is not the main selling point.

4) Encryption

In real systems, **TLS uses asymmetric crypto to exchange keys and symmetric crypto for bulk data**. That is a strong interview point.

5) Python scripts

For network automation, subprocess is usually a better choice than os.system. If you say os.system in an interview, they may ask why you did not use something safer.

How to answer in interviews

For every networking question, use this order:

Definition → Use case → Example → Verification / command → 1-line summary

That is the structure that sounds clean and confident.

1) Hub vs Switch vs Router

What it is

- **Hub**: sends data to all ports.
- **Switch**: sends data to the correct port using MAC address table.
- **Router**: connects different networks using IP addresses.

How to say it in interview

“Hub works at Layer 1 and broadcasts everything. Switch works mostly at Layer 2 and forwards frames based on MAC addresses. Router works at Layer 3 and forwards packets between different IP networks.”

Example

- Hub: one PC sends data, every device receives it.
- Switch: laptop sends data to printer, only printer gets it.
- Router: your home network reaches the internet through the router.

Useful command thinking

On a switch:

```
show mac address-table  
show vlan brief
```

On a router:

```
show ip route  
show ip interface brief
```

Interview trap

If someone asks “Can a switch route?” say:

“Yes, a Layer 3 switch can do routing, but a basic Layer 2 switch cannot.”

2) TCP/IP model vs OSI model

What it is

TCP/IP has **4 layers**. OSI has **7 layers**.

Mapping

- **TCP/IP Link** → OSI Physical + Data Link
- **TCP/IP Internet** → OSI Network
- **TCP/IP Transport** → OSI Transport
- **TCP/IP Application** → OSI Session + Presentation + Application

How to say it in interview

“OSI is a conceptual 7-layer reference model. TCP/IP is the real-world protocol stack used on the internet. TCP/IP compresses OSI layers into 4 layers.”

Example

When you open a website:

- Application: HTTP/HTTPS
- Transport: TCP
- Internet: IP
- Link: Ethernet/Wi-Fi

Commands to remember

ping
tracert
nslookup
ipconfig /all # Windows
ifconfig / ip a # Linux

Important add-on

Know **encapsulation and decapsulation**. That is interview gold.

3) ARP

What it is

ARP maps an **IP address to a MAC address** in a local network.

How to say it in interview

“ARP is used when a device knows the destination IP but needs the MAC address to send the frame on the local LAN.”

Example

Your laptop wants to send data to 192.168.1.5.
It first checks ARP cache. If missing, it sends an ARP broadcast:
“Who has this IP?”
The device replies with its MAC.

Commands

arp -a
ip neigh

Interview follow-up

Know **ARP cache**, **gratuitous ARP**, and **ARP spoofing**.

Security note

ARP has no built-in authentication, so it can be attacked.

4) OSPF areas

What it is

OSPF splits a large network into areas to reduce routing overhead and improve scalability.

Correct area types

- **Backbone area (Area 0):** central area all others connect to
- **Regular/normal area:** standard area
- **Stub area:** blocks external routes
- **Totally stubby area:** blocks more routes than stub
- **NSSA:** allows limited external route injection

How to say it in interview

“OSPF area design reduces LSA flooding and improves scalability. Area 0 is mandatory backbone. Other areas connect through ABRs.”

Example

A big company with 500 routers may split branches into areas so every router does not learn every route in full detail.

Must-know terms

- **LSA:** link-state advertisement
- **LSDB:** link-state database
- **ABR:** area border router
- **ASBR:** autonomous system boundary router

Hard part

If this feels confusing, learn it by drawing a 3-area topology and watching route propagation in GNS3/EVE-NG. That is the real way.

5) MPLS

What it is

MPLS forwards traffic using **labels** instead of doing a full IP lookup at every hop.

How to say it in interview

“MPLS sits between Layer 2 and Layer 3 conceptually. The ingress router labels the packet, core routers switch based on the label, and egress router removes it.”

Example

Used by service providers to build:

- VPNs
- traffic-engineered paths
- scalable WANS

Honest correction

Do not oversell speed. The benefit is more about **predictable forwarding and network engineering** than raw packet speed.

Keywords to mention

- Label Switched Path (LSP)
 - ingress router
 - transit router
 - egress router
-

6) Symmetric vs Asymmetric encryption

What it is

- **Symmetric:** same key encrypts and decrypts
- **Asymmetric:** public key encrypts or verifies, private key decrypts or signs

How to say it in interview

“Symmetric crypto is faster and used for bulk data. Asymmetric crypto solves key exchange and identity problems, but it is slower.”

Example

- Symmetric: AES for file encryption
- Asymmetric: RSA or ECC for key exchange / signatures

Strong interview point

In HTTPS/TLS:

- asymmetric crypto helps establish trust
- symmetric crypto protects the actual data stream

Security angle

If the key is stolen in symmetric crypto, the whole communication is exposed.

7) VLAN on Cisco switch

What it is

VLAN creates logical segmentation on the same physical switch.

How to say it in interview

“VLAN separates broadcast domains. It improves security, reduces broadcast traffic, and gives better network organization.”

Commands

```
enable
configure terminal
vlan 10
name Sales
interface fastEthernet 0/1
switchport mode access
switchport access vlan 10
end
show vlan brief
copy running-config startup-config
```

Example

- VLAN 10 = Sales
- VLAN 20 = HR

Even if they use the same switch, Sales and HR stay separated unless routed.

Important add-on

Also know:

```
switchport mode trunk
switchport trunk allowed vlan 10,20
```

because trunking is a very common follow-up.

8) Troubleshooting slow network connectivity

What it is

This is not a single answer. It is a process.

How to say it in interview

“I troubleshoot from bottom to top: physical, link layer, IP layer, DNS, application, then WAN or ISP.”

Clean troubleshooting flow

1. Check whether the problem is one user or many users
2. Check cables, link lights, NIC status
3. Check IP, gateway, DNS
4. Ping local gateway
5. Ping external IP
6. Test DNS resolution
7. Check interface errors and drops
8. Check bandwidth utilization and rogue traffic
9. Check firewall, QoS, and WAN link health

Commands

Linux:

```
ip a
ip route
ping 8.8.8.8
traceroute google.com
nslookup google.com
ss -tulnp
tcpdump
```

Windows:

```
ipconfig /all
ping
tracert
nslookup
netstat -an
```

Cisco:

```
show interfaces
show ip route
show arp
show logging
show processes cpu
```

Interview trap

Do not say “restart everything.” That is not troubleshooting. That is panic.

9) Python script to find failed SSH login attempts

What it is

Parse auth logs and search for failed login patterns.

Better interview answer

“Read the authentication log line by line, filter lines containing failed SSH login patterns, and count or store them.”

Sample script

```
import re

pattern = re.compile(r"failed password|invalid user", re.IGNORECASE)

with open("/var/log/auth.log", "r", errors="ignore") as f:
    for line in f:
        if pattern.search(line):
            print(line.strip())
```

Example

It will catch lines like:

- Failed password for invalid user root
- Invalid user admin

Stronger point

In production, you would also:

- count attempts by IP
- write alerts
- use log rotation support

10) Backup network configs from multiple devices

What it is

Automate SSH login to devices, run show commands, and save output.

Better interview answer

“This is basic network automation. Use SSH libraries, loop through a device list, run show running-config, save backups with timestamps, and schedule it.”

Example using Paramiko

```
import paramiko
from datetime import datetime

devices = ["192.168.1.1", "192.168.1.2"]

for ip in devices:
    ssh = paramiko.SSHClient()
    ssh.set_missing_host_key_policy(paramiko.AutoAddPolicy())
    ssh.connect(ip, username="admin", password="admin123")

    stdin, stdout, stderr = ssh.exec_command("show running-config")
    config = stdout.read().decode()

    filename = f'{ip}_{datetime.now().strftime('%Y%m%d_%H%M%S')}.txt'
    with open(filename, "w") as f:
        f.write(config)

    ssh.close()
```

Interview improvement

Use:

- cron on Linux
- secure credential storage
- error handling
- logging

Critical note

Never hardcode real passwords in a real system.

11) Subnet mask from CIDR

What it is

CIDR /24 means 24 network bits and 8 host bits.

How to say it in interview

“CIDR tells how many bits are fixed for the network. A /24 becomes 255.255.255.0.”

Quick examples

- /8 → 255.0.0.0
- /16 → 255.255.0.0
- /24 → 255.255.255.0
- /30 → 255.255.255.252

Python example

```
def cidr_to_mask(cidr):
    prefix = int(cidr.split('/')[1])
    mask = (0xffffffff << (32 - prefix)) & 0xffffffff
    return ".".join(str((mask >> (8 * i)) & 255) for i in [3, 2, 1, 0])
```

```
print(cidr_to_mask("192.168.1.10/24"))
```

Interview trap

You should also know:

- network address
- broadcast address
- usable host range

That is more important than the conversion code itself.

12) Network topology for 3 branches + central data center

What it is

This is a design question. They want architecture thinking, not poetry.

How to say it in interview

"I would use a hub-and-spoke design with the data center as the core, branches connecting over redundant WAN links, and routing/security policies enforced centrally."

Good design

- Data center has core switches, firewalls, servers
- Each branch has access switch, edge router/firewall
- Site-to-site VPN or private WAN
- Redundant links for failover
- Separate VLANs/subnets for users, voice, servers, guest

Example

Branch 1 users access ERP in data center via secured tunnel.
If one WAN link fails, traffic fails over to backup link.

Extra interview points

Mention:

- HA firewalls
 - load balancing
 - route summarization
 - failover time
 - monitoring
-

13) Data redundancy for financial provider

What it is

Financial systems need high availability, disaster recovery, and data integrity.

How to say it in interview

“I would use RAID for disk-level redundancy, replication for site-level redundancy, backup for recovery, and redundant network/ISP links for availability.”

Strong design ideas

- RAID 1/10 for critical storage
- real-time replication
- backup in different location
- active-active or active-passive DR site
- dual power and dual ISPs
- load balancers
- snapshot + backup policy

Interview correction

RAID is not backup. That mistake is common and bad.
RAID protects against disk failure, not accidental deletion or ransomware.

14) Wireless network design for open office

What it is

Wi-Fi design is about coverage, capacity, and interference.

How to say it in interview

“I would do a site survey first, then place APs based on capacity and coverage, use proper channel planning, and secure it with enterprise authentication.”

Example

- 2.4 GHz channels: only 1, 6, 11
- Prefer 5 GHz / 6 GHz for performance
- Use WPA3 or WPA2-Enterprise
- Separate guest Wi-Fi

Important points

- wall material matters
- AP placement matters
- too many APs can also hurt performance
- band steering and power tuning matter

Interview trap

Do not say “just add more APs.” That is a stupid answer if not designed carefully.

15) List devices on the same network with Python

What it is

Use ARP scanning to discover active hosts in the LAN.

How to say it in interview

“Send ARP requests across the subnet and collect responses. Any device that replies is on the local network and reachable at Layer 2.”

Example using Scapy

```
from scapy.all import ARP, Ether, srp

target_ip = "192.168.1.0/24"
arp = ARP(pdst=target_ip)
ether = Ether(dst="ff:ff:ff:ff:ff:ff")
packet = ether / arp

result = srp(packet, timeout=2, verbose=False)[0]

for sent, received in result:
    print(received.psrc, received.hwsrc)
```

Important note

This works on the local subnet, not across routed networks.

16) Simple socket server in Python

What it is

A socket server listens for client connections and responds.

How to say it in interview

“A socket server binds to an IP and port, listens for connections, accepts a client, and exchanges data using TCP or UDP.”

Example

```
import socket

server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server.bind(("127.0.0.1", 9999))
server.listen(5)

print("Server listening...")

conn, addr = server.accept()
print("Connected by", addr)
conn.sendall(b"Welcome!")
conn.close()
server.close()
```

Interview follow-up

Know:

- TCP vs UDP
- blocking vs non-blocking sockets
- bind, listen, accept, connect

17) Ping host and return true/false

What it is

A simple reachability check.

Better interview answer

“Use subprocess instead of os.system, because it is cleaner and lets you capture return code properly.”

Example

```
import subprocess
import platform

def is_host_up(host):
    count_flag = "-n" if platform.system().lower() == "windows" else "-c"
    result = subprocess.run(["ping", count_flag, "1", host],
                           stdout=subprocess.DEVNULL,
                           stderr=subprocess.DEVNULL)
    return result.returncode == 0

print(is_host_up("8.8.8.8"))
```

Interview point

Ping uses ICMP, so a host may be up even if ping is blocked by firewall.

18) DoS attack and prevention

What it is

A DoS attack tries to make a service unavailable by overwhelming it.

How to say it in interview

“DoS attacks can target bandwidth, protocol state, or application resources. Prevention needs rate limiting, filtering, redundancy, and upstream protection.”

Types

- volumetric: flood bandwidth
- protocol: SYN flood
- application: HTTP flood

Prevention

- firewalls
- rate limiting
- SYN cookies
- load balancers
- DDoS protection service
- WAF
- IDS/IPS
- autoscaling
- ISP blackholing or scrubbing

Important correction

A simple firewall alone is not enough against serious DDoS.

19) QoS parameters

What it is

QoS prioritizes important traffic when the network is congested.

How to say it in interview

“QoS is about classification, marking, queuing, shaping, and policing so real-time traffic like voice gets priority over bulk traffic like downloads.”

Example

- VoIP gets high priority
- file backups get lower priority

Important terms

- DSCP
- latency
- jitter
- packet loss
- shaping vs policing
- queuing

Interview trap

QoS does not create bandwidth. It only decides who gets priority when bandwidth is limited.

20) Routing protocol in a converged network

What it is

Routing protocols dynamically find paths between networks.

How to say it in interview

“In a converged network carrying voice, video, and data, routing protocols maintain reachable paths, react to failures, and support scalability. They help the network converge quickly after a change.”

Example

If a WAN link fails, OSPF or EIGRP finds another path.

Important points

- convergence speed
- metric
- loop avoidance
- route advertisement
- scalability
- failover

Strong add-on

Know the basic difference between:

- **static routing**
 - **dynamic routing**
 - **distance vector**
 - **link state**
 - **path vector**
-

What is missing and should be covered for interviews

These topics are more important than some of the Python scripting questions if you are going into networking interviews.

Must know basics

- IP addressing
- subnetting
- CIDR
- default gateway
- DNS
- DHCP
- NAT/PAT
- TCP vs UDP
- 3-way handshake
- ports and services
- ICMP
- traceroute
- ARP
- MAC address

Must know switching

- VLAN
- trunking
- access port
- STP/RSTP
- MAC table

- broadcast domain
- collision domain

Must know routing

- static vs dynamic routing
- OSPF
- EIGRP
- BGP basics
- route redistribution
- route summarization

Must know security

- firewall
- ACL
- VPN
- SSL/TLS
- DoS/DDoS
- AAA
- 802.1X
- spoofing

Must know operations

- ping
- traceroute
- nslookup
- netstat / ss
- tcpdump
- Wireshark
- show running-config
- show interfaces
- show ip route

Hard topics and how to learn them properly

1) Subnetting

Do not memorize only formulas. Drill it daily until you can solve it fast.

Learn:

- network address
- broadcast
- host count
- first/last usable IP
- VLSM

2) OSPF

Draw topology by hand. Then practice:

- adjacency formation
- area types
- ABR/ASBR
- LSAs
- DR/BDR

3) VLAN and trunking

You must understand:

- access vs trunk port
- native VLAN
- allowed VLAN list
- inter-VLAN routing

4) QoS

Focus on:

- classification
- marking
- queuing
- shaping
- policing

5) MPLS

Do not start with vendor marketing. Learn:

- label swapping
- LSP
- PE/P/CE roles
- why service providers use it

Best way to learn

Use:

- Packet Tracer for basics
- GNS3 or EVE-NG for deeper practice
- Wireshark for packet visibility
- show commands on Cisco-style labs

Interview speaking style that works

Use short, sharp answers like this:

“At a high level, VLAN separates broadcast domains logically. For example, Sales and HR can be on the same switch but isolated through different VLANs. On Cisco, I create the VLAN, assign access ports, and verify with show vlan brief.”

That sounds like someone who has actually worked with the thing.

The real mistake to avoid

Do not recite textbook definitions like a robot.
Instead, speak like this:

1. what it is
2. where it is used
3. why it matters
4. one real example
5. one command to verify it

That is the difference between “studied networking” and “can survive an interview.”

If you want the next step, I'll turn this into a **clean 1-page interview cheat sheet** or a **mock interview Q&A set with answers you can memorize**.